Total No. of Questions : 8]

SEAT No. :

PC-2845

[Total No. of Pages : 4

[6352]-69

S.E. (A.I&M.L)

DATA STRUCTURES & ALGORITHMS

(2019 Pattern) (Semester - III) (218542)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates :

1) ***Attempt question Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.***
2) ***Draw neat & labelled diagrams if necessary.***
3) ***Assume suitable data if necessary.***
4) ***Figures to the right side indicate frill marks.***

Q1) a) Convert following Infix Expression to Postfix and evaluate using stack **[6]**

(A/B–C+D*E)

A = 30 B = 2 C = 5 D = 10 E = 6

b) What is queue? How they are represented in memory? Write a pseudocode to implement insert & delete operation in a linear queue using array. **[6]**

c) What is priority queue? How insertion & deletion operation is performed? Explain with example. **[6]**

OR

Q2) a) Convert following expression into postfix form. (Show all intermediate steps) **[6]**

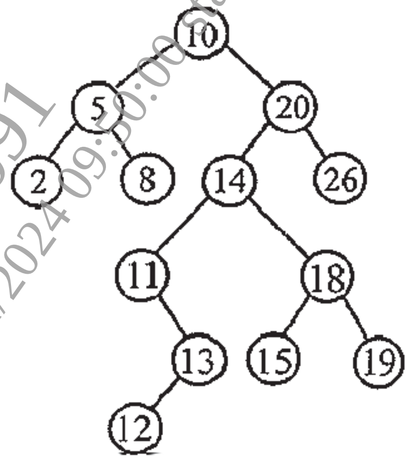
((A + B) - C * (D / E)) + F

b) Explain the concept of linear queue and circular queue.Give the advantages of circular queue over linear queue. Write C/C++ code to implement enqueue & degueue operation on circular queue. **[6]**

c) What is linked stack? Write a C/C++ code to implement insertion & deletion operation on it. **[6]**

P.T.O.

Q3) a) In a given BST write the output of inorder, preorder, postorder, level wise traversals. Also draw mirror image. **[6]**



b) Discuss algorithm for recursive and non recursive inorder traversal. **[6]**

c) Construct BST from below data **[6]**

lnorder - 17, 32, 44, 48, 50, 62, 78, 88

Postorder - 32, 17, 48, 62, 50, 88, 78, 44

OR

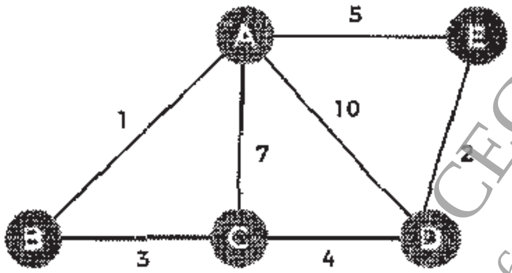
Q4) a) Construct the expression tree from the following prefix expression using stack. **[6]**

/, * , – , 4, 7, 9, 3

b) State and explain algorithm to display binary tree level wise with the help of sample tree. **[6]**

c) Draw and explain how Threaded Binary Tree is efficient than Binary tree. **[6]**

Q5) a) In the following graph, consider 'A' is the source. Apply Kruskal's algorithm to get minimum spanning tree. **[6]**



b) Find the Optimal Binary Search Tree for the : **[6]**

Identifier set {al, a2, a3} = {do, if, while} Where n = 3 and Probabilities of successful search as {pl, p2, p3) = {0.5, 0.1, 0.05) and Probability of unsuccessful search as {q0, q1, q2, q3) = {0.15, 0.1, 0.05, 0.05)

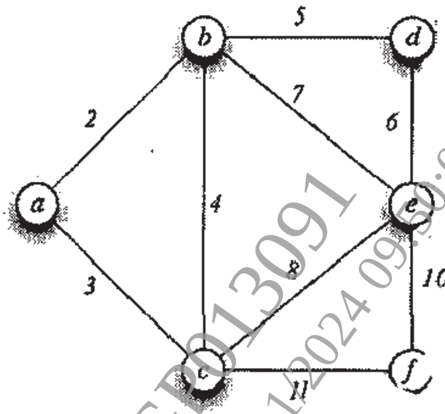
c) Explain the heap data structure &it's types with the help of sample heap tree. **[5]**

OR

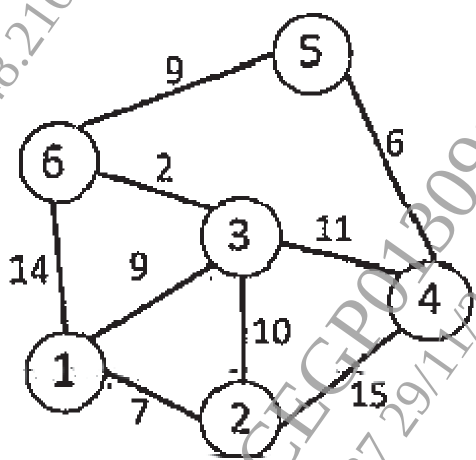
Q6) a) Construct AVL tree for following set of keys :

A, Z, B, Y, C, X, D, E, V, F, M, R. Show the balance factor of all the nodes & name the' type of rotation used. **[6]**

b) Show the output of Breadth First Search using stack. Use 'd' as a starting node **[6]**



c) If '5' is the source, find the shortest path from source to all vertices, using Dijkstra's algorithm. Show answer step by step. **[5]**



Q7) a) Create the hash table using Linear Probing. **[6]**

Table size : 15

Data : 30, 61, 46, 77, 33, 93, 105, 70, 1

Hash function: data% table size

b) Explain with example characteristics of good hash function. **[6]**

c) Discuss an algorithm to delete a record from the sequential file. **[5]**

OR

Q8) a) Discuss chaining with replacement technique to handle collision. Mention suitable example. **[6]**

b) Compare sequential, Index sequential and direct access file organization. **[6]**

c) Give an algorithm to merge two sequential files into third new sequential file. **[5]**

⌘⌘⌘